

Project Requirement Document

AI Customer Support Automation — Capstone Project (Option 1)

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Phase	Requirement Analysis
Document	Project-Requirement-Document-CustomerSupport.pdf

1. Problem Statement

Petalnex delivers custom software, AI, and automation solutions to clients on an ongoing project basis. Once a solution is deployed — a workflow, chatbot, integration, or web application — clients need a reliable channel to report issues, request changes, or check on progress. Handling these requests manually makes it hard to triage urgency, track resolution status, or maintain a consistent record of what was asked and how it was resolved.

2. Current Process & Pain Points

Current process: Client support requests arrive via email or informal messages, with no central intake point. A team member manually reads each request, decides what kind of issue it is, searches past project context if needed, and responds or escalates as they see fit.

Pain points:

- No central intake point — requests arrive through multiple informal channels
- No consistent way to classify request type (technical issue, integration failure, project status, bug, feature request, billing, general inquiry)
- Urgency isn't assessed consistently — a production-breaking bug and a routine status question can receive equal or misallocated attention
- No sentiment signal to flag frustrated or at-risk clients before they escalate further
- Responding often requires manually searching past project documentation for context
- No structured ticket record — requests can be lost, duplicated, or left without a clear resolution trail
- Escalation to the right team member (developer, project manager, billing) happens ad hoc

3. Proposed Solution Outline

Trigger & Input: A Form Trigger (or webhook) captures the client's inquiry: name, email, associated project, and message.

AI Classification: Google Gemini classifies each inquiry into one of seven categories: Technical Issue, Integration/API, Project Status, Bug Report, Feature Request, Billing, or General Inquiry.

Priority Detection: Rule-based checks combined with AI signal (e.g. production-down language, explicit deadlines, repeated contact) assign a priority of High, Medium, or Low.

Sentiment Analysis: Gemini reads the tone of the message to flag frustrated or at-risk clients, feeding into the overall priority score.

Knowledge-Base Search: A RAG pipeline over Petalnex's project documentation and past resolved tickets (Supabase with pgvector) surfaces relevant context for the specific client and project.

Suggested Response: Gemini drafts a reply grounded in the knowledge-base results, for human review — never auto-sent for anything beyond a general inquiry.

Ticket Creation: Each inquiry becomes a Supabase record with category, priority, sentiment, status (open / in-progress / resolved), and a full audit trail.

Human Escalation: Low-confidence classifications, high-priority or negative-sentiment tickets, and anything beyond a general inquiry are routed to a human before any response is sent — human-in-the-loop by design, consistent with the recruitment automation project.

Notifications: An acknowledgement email is sent to the client on intake; an internal notification routes to the right team (developer, project manager, or billing) based on category.

Data storage: Supabase Postgres holds ticket records, classifications, scores, statuses, and timestamps for a complete audit trail.

Error handling & security: Structured validation at each stage, retry logic for LLM/API calls, and logging of failures for review. Credentials are stored securely in n8n's credential store; no sensitive client data is exposed in workflow logs or automated emails.

4. Success Criteria

- Every inquiry is classified, prioritized, and stored with no manual triage required for routine cases
- High-priority or negative-sentiment tickets are reliably flagged and escalated to a human
- Suggested responses are grounded in real project context via the knowledge base, not generic text
- A full audit trail exists for every ticket: category, priority, sentiment, suggested response, human decision, and resolution
- The workflow handles normal, edge, and failure cases (missing fields, malformed AI output, failed knowledge-base search, failed email delivery) without breaking